PHYSICAL CHEMISTRY

GUIDED REVISION PLAN_SCORE ADVANCED [SOLUTION]

Topic : CHEMICAL KINETICS & RADIOACTIVITY (Question Bank # 07)

SINGLE CORRECT

1. (A)

Sol. $xA \longrightarrow yB;\ \frac{1}{x}\left(\frac{-d[A]}{dt}\right)=\frac{1}{y}\left(\frac{d[B]}{dt}\right)\ -\frac{d[A]}{dt}=\frac{x}{y}\left[\frac{d[B]}{dt}\right]$

$$\log\left(\frac{-d[A]}{dt}\right)=\log\left(\frac{d[B]}{dt}\right)+\log\left(\frac{x}{y}\right)$$

$$\text{Comparing with : } \log\left(\frac{-d[A]}{dt}\right)=\log\left(\frac{d[B]}{dt}\right)+0.3$$

$$\text{we get } \log\left(\frac{x}{y}\right)=0.3=\log 2\Rightarrow \frac{x}{y}=\frac{2}{1}$$

2. (D)

Sol. Rate = k [ester] $[H^+]$ {Note that H^+ is catalyst and its concentration will be constant}

rate = (0.01×0.05) $[ester]^1$

Pseudo - first order : $k_{eff} = 0.01 \times 0.05\ hr^{-1}$

$$t_{1/2}=\frac{0.693}{0.01\times 0.05}=1386\,hr$$

3. (C)

Sol. The slope of $\log(a_0 - x)$ vs time graph for 1st order $=\frac{-k}{2.303}$; from graph slope $=\frac{-2}{10}$

$$\frac{-k}{2.303}=\frac{-2}{10}\Rightarrow k=0.4606\,min^{-1}=\frac{0.4606}{60}sec^{-1}$$

$k = 7.7 \times 10^{-3}\ s^{-1}$

4. (A)

Sol. Let : rate = $k[A]^n$ (n = order)

rate = change in conc. per unit time

$$\frac{0.01\times 0.02}{1\,milli\,sec}=k(0.02)^n \quad ...(1)$$

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$$\frac{0.01\times 0.04}{0.25\,\text{milli sec}} = k(0.04)^n \qquad ...(2)$$

$$\frac{(1)}{(2)} \Rightarrow \frac{0.02\times 0.25}{0.04} = \left(\frac{0.02}{0.04}\right)^n \Rightarrow n = 3$$

5. (D)

Sol. $A \rightleftharpoons nB$

$t = 0 \quad P_0 \quad 0$

$t = t \quad P_0 - x \quad nx$

$P_0 - x = P_t = 1 - e^{-kt}$

$x = P_0 (1 - e^{-kt})$

$P_{Total} = P_0 - x + nx$

$P_{Total} = P_0 + (n-1)\,x$

$P_T = P_0 + (n-1)\,P_0 (1 - e^{-kt})$

$P_{Total} = P_0 [\,n + (1-n)\,e^{-kt}]$

6. (A)

Sol. When x was living let C^{14} atmosphere $= C_0$

25% increase will increase C^{14} to $= \frac{5C_0}{4}$

At present C^{14} conc. in fossil $= \frac{1}{5}\times\frac{5Co}{4} = \frac{Co}{4}$

$$t_{age} = \frac{6000}{\ell n\,2}\ell n\left(\frac{Co}{Co/4}\right) = 12000\ \text{years}$$

$= 1.2 \times 10^4$ years

7. (D)

Sol. Activation energy $(E_a) = RT^2\left(\frac{d}{dT}(\ln k)\right)$

$k = 30\,e^{\left(-\frac{40}{T}+\frac{300}{T^2}\right)}$ {Non-arrhenious behaviour}

$$\ell n\,k = \ell n\,30 - \frac{40}{T} + \frac{300}{T^2}$$

$$\frac{d}{dT}(\ell n\,k) = \left(0 + \frac{40}{T^2} - \frac{600}{T^3}\right) = \left(\frac{40}{T^2} - \frac{600}{T^3}\right)$$

$$E_a = RT^2\left(\frac{d}{dT}(\ell n\,k)\right) = \left(40R - \frac{600R}{T}\right)$$

$$\text{At } 300\text{ k}: E_a = 40R - \frac{600}{300}R = 38R$$

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8. (C)

Sol. $^{211}_{83}\mathrm{Bi(s)} \longrightarrow {}^{207}_{81}\mathrm{T\ell(s)} + {}^{4}_{2}\mathrm{He(g)} \longrightarrow \frac{30}{16}\text{mole}$

$t = 0 \qquad 2\,\text{mole}$

$t = 520\ \text{sec} \qquad \frac{2}{16}\text{mole}$

$(t_{1/2} = 130\,\text{sec})$

$$P = \frac{n_{He} \times RT}{V} = \frac{\frac{30}{16} \times 0.0821 \times 300}{100}$$

$P = 0.4681$ atm.

MULTIPLE CORRECT

9. (A,B,C)

Sol. For elementary reactions : stoichiometry $\rightarrow$ order

For zero-order : rate = k

Zero - order rate does not depend on concentration

Zero-order is always a complex reaction.

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10. (A,B,C)

Sol.

Reaction - I	Reaction - II
E_a	$2E_a$
$k'_1 = Ae^{-\frac{E_a}{RT'}}$	
$k_1 = Ae^{-\frac{E_a}{RT}}$	$\frac{k'_2}{k_2} = e^{\frac{+2E_a}{R}\left[\frac{1}{T}-\frac{1}{T'}\right]}$
$\frac{k'_1}{k_1} = e^{\frac{E_a}{R}\left[\frac{1}{T}-\frac{1}{T'}\right]}$	$\frac{k'_2}{k_2} > 1$
$\frac{k'_1}{k_1} > 1$	

$$\frac{k'_1}{k_1} < \frac{k'_2}{k_2} \text{ and } \frac{k'_2}{k_2} = \left(\frac{k'_1}{k_1}\right)^2$$

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11. (A,C)

Sol. Specific activity is activity per unit mass which is greater for U^{235} than $^{235}U_3O_8$

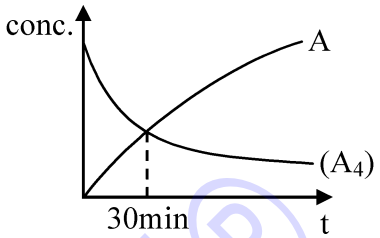
Decay constant (λ) does not depend on any factors. Every radioactive isotopic hasit's unique 'λ' irrespective of conditions in which it is present.

Radioactive disintegrations do not depend on temperature or any external factors.

12. (A,B,D)

Sol.

	$A_4 \longrightarrow$	$4A$
$t = 0$	a	0
$t = 30$ min	$a - x$	$4x$



At 30 min: $a - x = 4x$

$x = a/5$ (20% complete)

$$k = \frac{2.303}{t}\log\left(\frac{a}{4a/5}\right) = \frac{2.303}{30}\log\left(\frac{5}{4}\right)$$

$$t_{½} = \frac{\ell n\, 2}{k} = 90.28\,\text{min}$$

Rate of 1st order decreases exponentially with time. The time for concentration of A_4 and A to become equal does not depend on initial conc.

For this particular reaction, it will be 30 min.

13. (A,D)

Sol. Rate of decay = λN

Rate of decay depends on : (i) Amount, (ii) λ (which is unique to each radioactive isotope)

14. (B)

Sol.

	$2NH_3 \longrightarrow N_2 + 3H_2$ (zero-order)		
$t = 0$	3 atm		
$t = t$	$3 - 2p$	p	$3p$

$$\text{rate} = \frac{0.1\,\text{atm}}{\text{sec}} = -\frac{1}{2}\frac{d\left(P_{NH_3}\right)}{dt} \Rightarrow \frac{-d\left(P_{NH_3}\right)}{dt} = 0.2\,\text{atm/sec}$$

15. (B)

Sol. At $t = 10$ sec ; $\dfrac{-d\left(P_{NH_3}\right)}{dt} = -\dfrac{\{(3-2p)-3\}\,\text{atm}}{10\,\text{sec}} = 0.2\,\text{atm/sec}$

solving : $p = 1$

$P_{Total} = 3 - 2p + p + 3p = 3 + 2p = 5$ atm.

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16. (B)

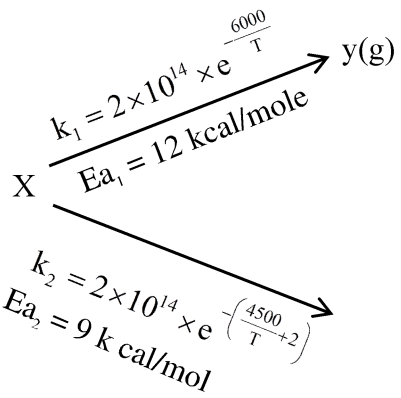
Sol. $E_{net} = \dfrac{k_1Ea_1 + k_2Ea_2}{(k_1 + k_2)}$

$10.5 = \dfrac{12k_1 + 9k_2}{(k_1 + k_2)}$

$\Rightarrow k_1 = k_2$

if $k_1 = k_2$ then : $e^{-\left(\frac{6000}{T}\right)} = e^{-\left(\frac{4500}{T}+2\right)}$

$\Rightarrow T = 750$ Kelvin



17. (D)

Sol. At 300 k : $k_1 = 2 \times 10^{14}\, e^{-20}$; $k_2 = 2 \times 10^{14}\, e^{-17}$

$k_1 < k_2$ (Z major product)

At 800 k : $k_1 = 2 \times 10^{14}\, e^{-7.5}$; $k_2 = 2\times10^{14}\, e^{-7.625}$

$k_1 > k_2$ (Y major product)

18. (B)

Sol. When temperature is increased, both rate of formation of Y and Z increases, but increase in Y will be more.

This is very evident from last question (Q : 17).

Hence, correct option is (B).

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SECTION-II (INTEGER)

1. (8)

Sol. $_ZX^A \xrightarrow{6\beta} {}_{(Z+6)}Y^A \xrightarrow{7\alpha} {}_{(Z+6-14)}P^{(A-28)}$

2. (7)

Sol. $A(+40) \xrightarrow{k=10^{-3}\,min^{-1}} 2B(+10) + C(-30)$

$t = 0$	$4a$		$3a$
$t = t$	$(4a - x)$	$2x$	$(3a + x)$

$+40(4a - x) + 10(2x) - 30(3a + x) = 0$

$x = \dfrac{7a}{5} \Rightarrow t = \dfrac{1}{k}\ell n\left(\dfrac{4a}{4a - x}\right)$

$t = \dfrac{1}{10^{-3}}\ell n\left(\dfrac{4a}{4a - \frac{7a}{5}}\right) \Rightarrow t = \dfrac{\ell n\left(\frac{20}{13}\right)}{10^{-3}} = 420\,min$

$t = \dfrac{420}{60} = 7\,hr$

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3. (4.00)

Sol. $CH_3COOC_2H_5 \xrightarrow[\text{Catalyst}]{[H^+]} CH_3COOH + C_2H_5OH$

t = 0	0.3		
t = 10 min	0.3 – x	x	x
t = ∞	0	0.3	0.3

Alkali reacts with H^+ and CH_3COOH both

let vol. of reaction mixture used = V

let conc. of alkali = C

$t = 0 \Rightarrow V[H^+] = C \times 25$ ...(1)

$t = \infty \Rightarrow V[H^+] + 0.3\ V = 40\ C$ ...(2)

from (1) & (2) $0.3\ V = 15\ C$ ...(3)

$t = 10$ min $V[H^+] + Vx = 28\ C$

$Vx = 3C$ ...(4)

from (3) and (4) $x = \dfrac{0.3}{5}$

$$k = \frac{1}{10}\ell n\left(\frac{0.3}{0.3-\frac{0.3}{5}}\right) = \frac{\ell n\left(5/4\right)}{10} = 0.02 \rightarrow \text{this is pseudo 1st order constant.}$$

$k_{pseudo} = k_{actual} \times [H^+]$

from above equation ; $[H^+] = 0.5$

$$k_{actual} = \frac{0.02}{0.5} = 0.04$$

Rate = $(0.04)\ (ester)^1\ (H^+)^1 = 0.04 \times 0.1 \times 0.1$

$= 4 \times 10^{-4}$

SECTION - IV : MATRIX MATCH

4. (A-P,R,S; B-P,R,S; C-Q,S; D-P,Q,S)

Sol. (A) Molecularity in number of molecules that react in elementary step. This number cannot be negative and is always a natural number.

(B) For a elementary step, the stoichiometric coeff (should not be fractions) give order. These stoichiometric coeff, should be natural numbers.

(C) Order in rate law can be negative, fraction.

(D) Rate constant (k) can never be negative and it only depends on temperature and presence of catalyst.

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